

Know

Water and Major Minerals – Guided Notes – NDFS 1020

Water is the most <u>essential</u> nutrient! We can only live for a few days without water, whereas we can survive for weeks or months without other nutrients.	List a population that is at-risk for dehydration: <u>older adults (kidneys absorb less water)</u>
Major minerals are needed in larger amounts (<u>100</u> mg or more) than trace minerals. <u>potassium</u> is the main intracellular cation. <u>sodium</u> is the main extracellular cation. <u>chloride</u> is the main extracellular anion.	Hyponatremia occurs when too much water is consumed, and <u>dilutes</u> the sodium concentration of the blood. Symptoms include: <u>nausea</u> /vomiting - Confusion - Convulsions <u>death</u>
<u>< 120 / < 80</u> is considered normal blood pressure 120-129/<80 is considered pre-hypertension >130/>=80 is considered hypertension	Blood Pressure can be regulated when the - Kidneys reabsorb more <u>water</u> - Kidneys reabsorb more <u>sodium</u> - Fluid Volume ~ Blood Volume ~ Blood Pressure <u>vasoconstriction</u> (Decreased diameter of the blood vessel = increased blood pressure)
The <u>DASH</u> (<u>dietary approaches to Stop hypertension</u>) Diet helps lower blood pressure. This diet is high in <u>calcium, magnesium</u> and potassium, but is low in <u>sodium</u> and <u>fat</u> .	In addition to bone health, calcium has the following roles: <u>nerve</u> transmission - Muscle contraction and heart beat <u>blood</u> clot formation - Cell metabolism - Maintaining healthy blood pressure and immune system - Reduces risk of heart disease, hypertension, insulin resistance, inflammation and hyperlipidemia
Most calcium in the body is stored in the <u>bones</u> .	
Low Blood Calcium Vitamin D and <u>parathyroid hormone</u> (PTH) stimulate a/n <u>↑</u> (circle one: <u>↑</u> ↓) in - Calcium released from bones - Calcium reabsorbed in the kidney - Absorption of calcium in the intestines	High Blood Calcium <u>Calcitonin</u> stimulates... (circle one: <u>↓</u>) calcium deposition in bone (circle one: <u>↑</u>) calcium reabsorption in the kidney (circle one: <u>↑</u>) absorption of calcium in the intestines
Supports Bone Health <u>weight bearing exercise</u> - Calcium - Vitamin D	Detrimental to Bone Health Alcohol
<u>calcium</u> , Magnesium, and <u>phosphorus</u> are involved in mineralization of bones and teeth.	It's recommended that adults consume <u>1000</u> mg of potassium daily. <u>men 70+, women 50+ = 1200</u> <u>adolescents = 1300</u>
<u>TF</u> Hyperkalemia is likely to occur from eating fresh fruits and vegetables.	Additional roles for Phosphorus include: - Bones and teeth - DNA & RNA - ATP <u>phospholipids</u>

CHAPTER 11 (Major Minerals)

Fxn = Function, Fx = Fracture, RBC = Red Blood Cells, N/V = Nausea/Vomiting, Synthesis / Synthesized = Creation of / Made, ↑ = Increases or Raises, ↓ = Decreases or Reduces → = Leads to, mg = milligrams, ug or mcg = micrograms, d = day, Rec'd = Recommendation

NUTRIENT	FUNCTION	DEFICIENCY	TOXICITY	SOURCES	COOL STUFF
SODIUM (Na)	Electrolyte (Needed for Fluid Balance) Affects Blood Volume & Blood Pressure (BP) Nerve Transmission, Muscle Contraction	HYPONATREMIA (Symptoms are Similar to Heat Stroke, Muscle Cramps, Apathy, Poor Appetite, Exhaustion, ↑ Body Temperature, N/V, Diarrhea)	HYPERNATREMIA (Edema or Swelling) HYPERTENSION (HTN) High Blood Pressure in Some Individuals, ↑'d Urinary Calcium Loss, UL = 2,300 mg/d	Processed Foods & Canned Goods Table Salt Luncheon Meat Snack Foods Condiments, Sauces	The DASH Diet (Dietary Approaches to Stop Hypertension) is ↓ in Sodium ↑ in Calcium, Potassium, Magnesium (Vegetables, Fruits, Whole Grains, Dairy)
CHLORIDE (Cl)	Electrolyte (Needed for Fluid Balance and Acid-Base Balance) Needed to Produce HCl (Hydrochloric Acid) for Digestion	Not Common (However, Convulsions Have Been Observed in Infants Who are Deficient in Chloride)	HYPERCHLOREMIA (Vomiting, ↑'d Risk of Hypertension or HTN) UL = 3,600 mg/d	Processed Foods Table Salt Salty Snack Foods Soy Sauce	Foods High in Sodium Only Taste Salty if Chloride is Also Present (NaCl)
POTASSIUM (K)	Electrolyte (Needed for Fluid Balance Nerve Transmission, Muscle Contraction Regulation of Heart Beat, Helps ↓ BP	HYPOKALEMIA (Muscle Weakness, Cramps, Irregular Heart Beat, Confusion, Glucose Intolerance)	HYPERKALEMIA (Muscle Weakness, Vomiting, Cessation of Heartbeat if Potassium is Injected or Too Much is Given Intravenously)	Fresh Fruits & Veggies (Bananas, Potatoes, Melons, Citrus Fruits) Milk & Dairy Products Meat, Legumes, Whole Grains	Anti-Hypertensive Rec'd = 4,700 mg/d (men and women)
CALCIUM (Ca)	Mineralization of Bones and Teeth Muscle Contraction & Relaxation, Nerve Transmission Blood Clotting, ↓ BP	OSTEOPOROSIS (Severe Loss of Bone, Usually Seen in Elderly) ↑'d Risk of HTN Stunted Growth in Children	Constipation, ↑'d Risk of Kidney Stones Can Interfere w/ Absorption of Other Nutrients (i.e. Iron) UL = 2,500 mg/d	Milk & Dairy Products (~ 300 mg/Serving) Small Fish w/ Bones, Tofu, Legumes, Some Green Leafy Veggies, Calcium-Fortified Foods	Anti-Hypertensive Rec'd = 1,000 mg/d (adults), 1,300 mg/d (teens), 1,200 mg/d (seniors)
PHOSPHORUS (P)	Mineralization of Bones and Teeth Acid-Base Balance Needed to Make ATP (Energy), Phospholipids (in Cell Membranes)	Not Common	Poor Bone Mineralization, ↑'d Calcium Loss from Bones UL = 4,000 mg/d	Milk & Dairy Products Eggs Meat, Fish, Poultry Processed Foods & Baked Good	Phosphorus is the Second Most Abundant Mineral in the Body (Calcium is the Most Abundant)
MAGNESIUM (Mg)	Mineralization of Bones and Teeth Protein Synthesis, Cofactor for Certain Enzymes, Nerve & Heart Fxn, Helps ↓ BP	Muscle Weakness, Poor Heart Function, Difficulty Swallowing, Growth Failure in Children, Convulsions or Hallucinations (in Extreme Cases)	Occurs Only with Supplementation or with Medication Diarrhea, Disrupted Acid-Base Balance, Dehydration	Nuts, Legumes Whole Grains Dark Green Veggies Seafood Chocolate or Cocoa	Anti-Hypertensive The More Magnesium You Eat, the Less You Absorb, Average Absorption Rate For Most Individuals = 50%

CHAPTER 11 (WATER)

NUTRIENT	FUNCTION	DEFICIENCY	TOXICITY	SOURCES	COOL STUFF
WATER (H2O)	Nutrient Transport, Waste Removal, Metabolic Reactions, Body Temperature Regulation, Moistens Tissue, Absorbs Shock, Protects Organs	DEHYDRATION (Thirst, Dry Mouth, Weakness, Exhaustion, ↓'d Urination or Dark Urine, Sunken Eyes, Dizziness, Confusion, Irritability, Delirium)	OVERHYDRATION or HYPONATREMIA (Symptoms Similar to Heat Stroke) Death is Possible w/ a Deficiency or Toxicity	Drinking Water (from Tap or Bottled) Non-Alcoholic Beverages (Milk, Juice) and Most Foods Metabolic Reactions	Hard Water has Minerals (Ca, Fe) and is Recommended Over Soft Water (↑ in Na), No Major Nutritional Difference Between Tap Water & Bottled Water

Water

- Body composition
 - 50-75% of the body is water, depending on age/sex/body composition
 - Young males = ~60%
 - Young females = less water, more fat
 - Newborns = ~75%
 - Older adults = ~45%
 - Lean muscle has more water than fat
- Functions of water
 - Major solvent
 - Participates in chemical reactions like digestion

TABLE 11.1 Functions of Water in the Body

Water

- Is a solvent
- Is a major component of blood, saliva, sweat, tears, mucus, and joint fluid
- Removes wastes
- Helps transport substances
- Lubricates tissues
- Regulates body temperature
- Helps digest foods
- Participates in many chemical reactions
- Helps maintain proper blood pH

- Membrane transport
 - Helps transport water soluble substances within and around cells (simple diffusion, selective permeability and osmosis)
 - Water moves from areas of less material to areas of more material (to balance out the concentrations, at which point diffusion stops)
- Intracellular vs extracellular
 - Intracellular
 - Water inside cells
 - About $\frac{2}{3}$ of the body's water
 - Extracellular
 - Water surrounding cells (interstitial, plasma in blood)
 - Having too few sodium ions can cause water to move into the cells, which will burst
 - Having too many sodium ions will cause water to move out of the cell, which will shrivel and die (edema)
 - The body regulates these concentrations with electrolytes (sodium, **potassium**, **phosphate**, chloride) known as ions. Bold = primarily for intracellular, underlines = extracellular
- Sources of water
 - Individual water needs are dependent on factors such as
 - Humidity
 - Environmental temp
 - Age
 - Health conditions
 - Physical activities

- Dietary choices
- Total water intake = all water from food or beverages
- AI for total water intake is ~11c for young females and ~15.5c for young males
 - This includes other drinks, most foods, and fruit especially!
- Metabolic water is a by product of metabolism, ranges from 1-2.5c a day based on activity levels

Hydration status

- Tell if someone is hydrated by looking at urine volume and color (with exceptions, some meds/foods/supplements mess with color)
- Signs of dehydration:
 - Rapid weight loss
 - Dry or sticky mouth
 - Low or no urine output
 - Dark urine
 - No tears
 - Sunken eyes
 - Confusion
 - Heart palpitations
 - Weakness
- 16 oz of water = 1 lb of body weight
- Losing 4% of body water weight = considerable loss of muscle strength and endurance
- Losing 7-10% of body weight = severe weakness
- 20% = coma and death are likely
- Older adults do not sense thirst as accurately as young adults, and are more likely to experience dehydration as their kidneys become less able to conserve water when intake is low
- Diuretics increase urine production (coffee, tea, energy drinks, soft drinks, basically just caffeine in general)
 - Note: these still have more water intake than they cause loss, so they are beneficial
 - Alcohol causes more water loss than you gain by drinking it, it contributes to dehydration
- Water intoxication can happen when an excessive amount of water is consumed quickly, or if the kidneys have difficulty filtering water from the blood.
 - Dilutes the sodium concentration in the blood (hyponatremia) and causes too much water to enter cells
 - Drowsiness, nausea, vomiting, confusion, inability to coordinate muscular movements, bizarre behavior, seizures, can lead to coma and death
 - Mostly happens in endurance athletes who drink a lot of plain water

Calcium

- Functions
 - Bone and teeth formation
 - Muscle contraction
 - Blood clot formation
 - Nerve pulse transmission
 - Cell metabolism
 - Maintaining blood pressure and immune function
- Food sources
 - Milk products, fortified cereals or drinks, broccoli, leafy greens
 - Plants are usually less bioavailable

TABLE 11.5 Calcium Content of Selected Foods

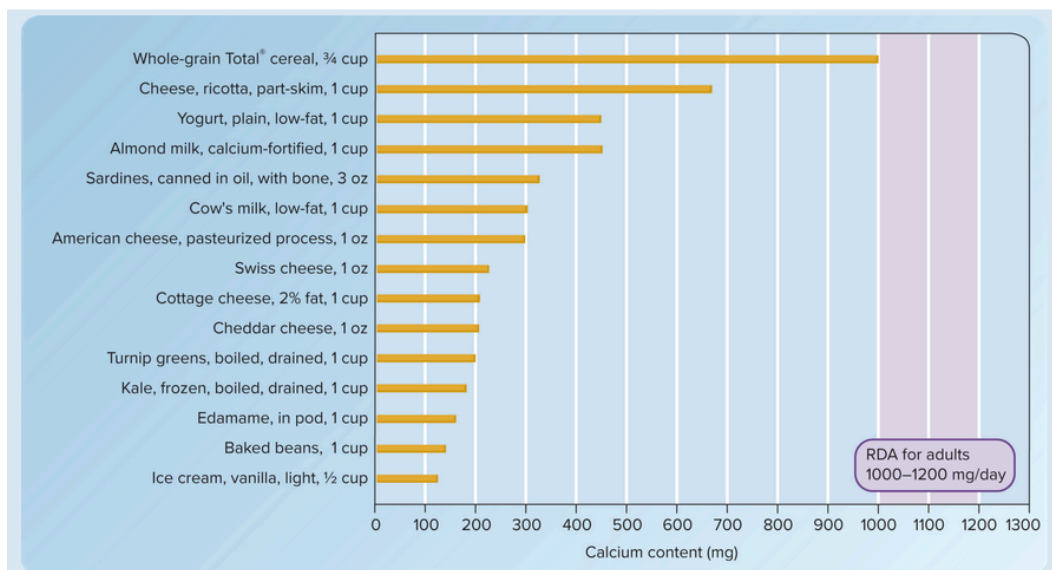


TABLE 11.6 Factors That Influence Calcium Absorption

Enhance Calcium Absorption	Decrease Calcium Absorption
Vitamin D (calcitriol)	Vitamin D deficiency
Stomach acid	Reduced stomach acid secretion
Lactose (only in infants)	Phytic and oxalic acid intake
	High fiber intake
	High phosphorus intake
	Chronic diarrhea

- Dietary adequacy

- RDA = 1000 mg
 - 1200 mg for men 70+ and women 50+
 - 1300 mg for adolescents
- Deficiency
 - Osteoporosis
 - Low bone mass and reduced bone structure
 - Increased risk of fractures
 - Fractures can heal in abnormally curved ways, especially in upper back

TABLE 11.7 Risk Factors for Osteoporosis

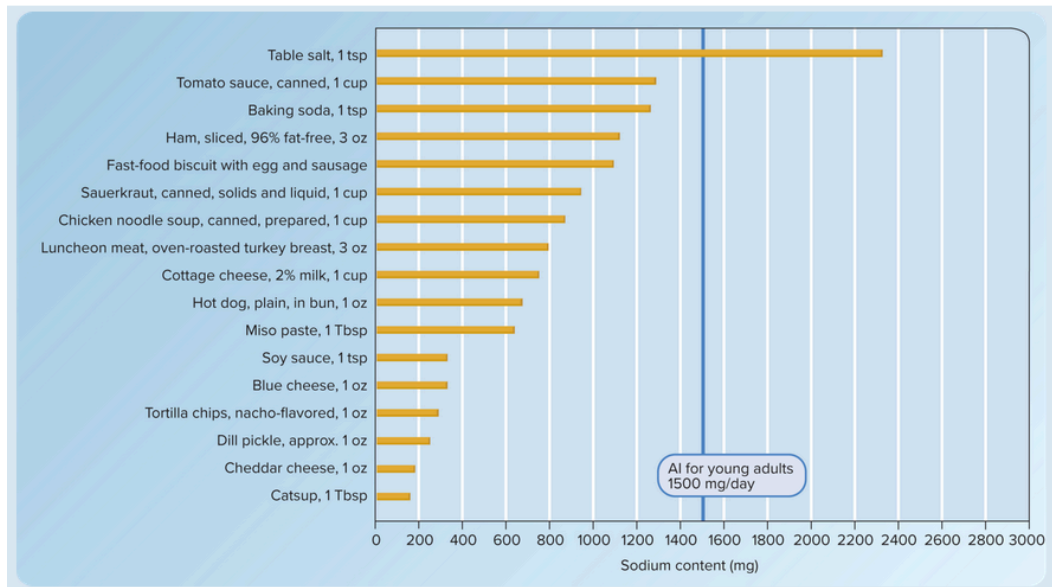
Nonmodifiable Risk Factors	Modifiable Risk Factors
Female	Having anorexia nervosa
Growing older	Having low estrogen levels (females) or low testosterone levels (males)
Having white or Asian ancestry	Following diets that contain inadequate amounts of calcium and vitamin D
Having a family history of osteoporosis	Using medications such as steroids or some types of anticonvulsants (antiseizure drugs)
Having a small, thin-boned body frame	Being physically inactive
	Smoking cigarettes
	Consuming excessive amounts of alcohol

- Toxicity
 - UL = 2000 - 2500 mg
 - Hypercalcemia
 - Higher than normal levels of calcium in the blood
 - Kidney stones
 - Bone pain
 - Muscle weakness
 - Fatigue
 - hypertension

Sodium

- Functions
 - Maintaining fluid balance
 - Transmission of nerve impulses
 - Muscle function
 - Transport of small molecules like glucose and amino acids into cells
- Food sources
 - Processed foods, salt as an additive, deli meats/cheeses, pickled foods

TABLE 11.10 Sodium Content of Selected Foods



- Dietary adequacy
 - RDA = 1500 mg (increases if you sweat a lot)
 - Deficiency
 - Hyponatremia
 - Weight loss
 - Headache
 - Nausea
 - Vomiting
 - Diarrhea
 - Fatigue
 - Muscle cramps
 - In serious cases leads to seizures, coma, and death
 - Toxicity
 - No formal UL, but recommended to be under 2300 mg
 - Increased risk of hypertension
 - Water retention
 - Increased blood volume and pressure

Chloride

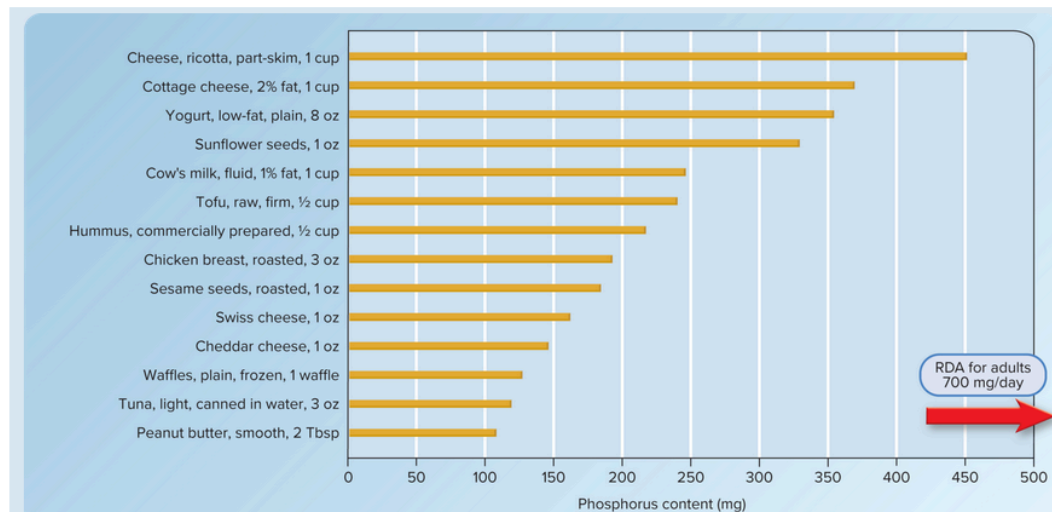
- Functions
 - Regulate fluid balance
 - HCL production
 - Acid base balance
 - Transmission of nerve impulses
 - Immune fighting response of white blood cells
- Food sources
 - Salt, seaweed, rye, tomatoes, celery, olives

- Dietary adequacy
 - AI = 2300 mg
 - Deficiency
 - Unlikely
 - Extreme vomiting
 - Loss of HCL in gastric juice, results in acid base and fluid imbalance
 - Diarrhea
 - Sweating
 - Fatigue
 - Loss of appetite
 - Toxicity
 - UL = 3600 mg
 - hypertension

Phosphorus

- Functions
 - Synthesize ATP and creatine phosphate for energy production
 - Synthesize phospholipids, DNA, RNA
 - Regulate acid-base balance
 - Activate many enzymes and hormones needed for energy metabolism
- Food sources
 - Dairy products, meat, cereal grains

TABLE 11.9 Phosphorus Content of Selected Foods



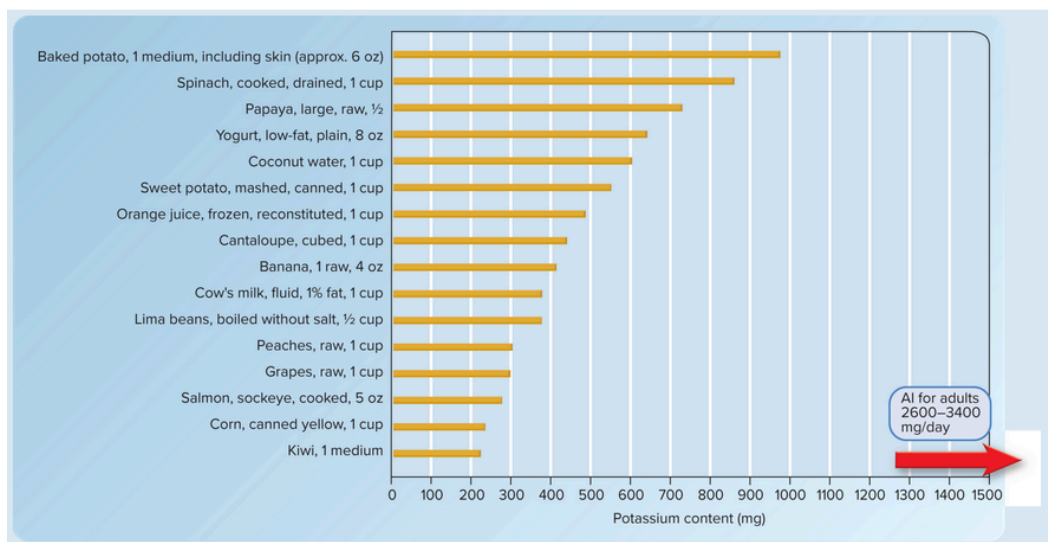
- Animal foods are more bioavailable (they have inorganic phosphorus)
- Dietary adequacy
 - RDA = 700 mg
 - Deficiency
 - Rare

- Bone mass loss
- Decreased bone mineralization and strength
- Rickets
- Poor appetite
- Weight loss
- Fatigue
- Muscle pain
- Weakened immune system
- Bone pain
- Toxicity
 - UL = 4000 mg
 - Hyperphosphatemia
 - Causes calcium-phosphate deposits in body tissues, including blood vessels

Potassium

- Functions
 - Maintain fluid balance
 - Associated with lower blood pressure values
 - Transmitting nerve impulses
 - Contracting muscles
 - Maintaining kidney function
- Food sources
 - Fresh fruit and veggies, cows milk, whole grains, dried beans, and meats

TABLE 11.13 Potassium Content of Selected Foods



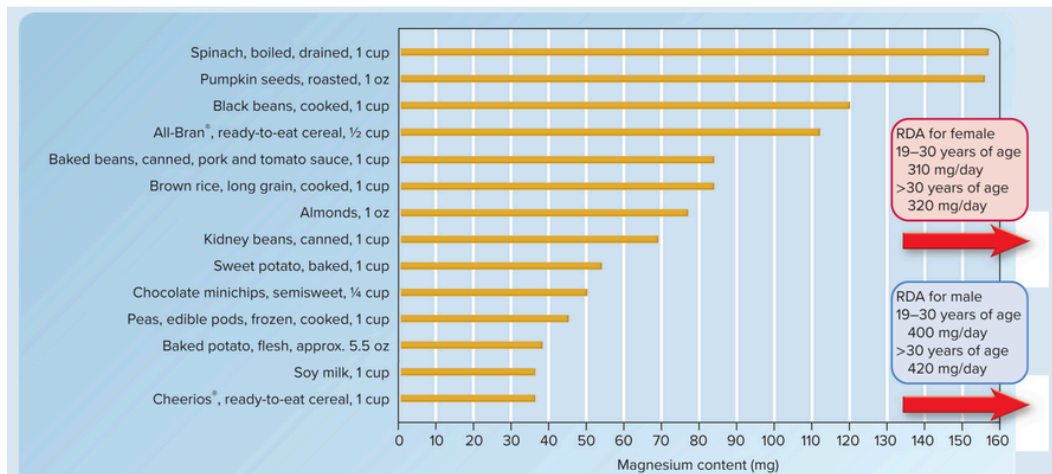
- Dietary adequacy
 - RDA = 2600 mg for females, 3400 mg for males
 - Deficiency

- Increased risk
 - Excessive sweating
 - Vomiting
 - Diarrhea
 - Kidney disease
- Hypokalemia
 - Loss of appetite
 - Muscle cramps
 - Confusion
 - Constipation
 - Increased urinary calcium excretion
- Toxicity
 - No UL
 - Supplements can cause hyperkalemia
 - Accumulates in blood and can cause heart to stop beating (used for lethal injections)

Magnesium

- Functions
 - Muscle and nerve function
 - Enzyme function
 - Energy production
 - DNA and protein synthesis
 - Blood pressure and glucose levels
 - Strong bones
 - Healthy immune system
- Food sources
 - Chlorophyll
 - Spinach, leafy greens, whole grains, beans, nuts, seeds, chocolate, cows milk, meats, hard tap water

TABLE 11.14 Magnesium Content of Selected Foods



- Dietary adequacy
 - RDA = 310- 420 mg
 - Deficiency
 - At risk
 - Alcoholics
 - Poorly controlled diabetes
 - Certain meds (diuretics)
 - Older adults (decreased absorption, increased excretion)
 - Irritability
 - Weakness
 - Loss of appetite
 - Muscle twitching
 - Severe cases
 - Rapid heartbeat
 - Inability to relax muscles
 - Disorientation
 - Hallucinations
 - Increased risk of osteoporosis
- Toxicity
 - UL = 350 mg, but from meds not food sources
 - Hypermagnesemia
 - Diarrhea
 - If person has kidney failure
 - Weakness
 - Nausea
 - Slowed breathing
 - Coma
 - death

Skim

Water balance and hydration

- Hydration = body's water status
- Average adult should consume about 2.5 quarts of water a day, as that is the amount that is lost through urine, exhaled air, feces, and perspiration daily
- Water intake should = water loss, which can be impacted by
 - Temperature
 - Humidity
 - Wind
 - Altitude
 - Fever
 - Vomiting
 - Diarrhea
 - Exercise habits
 - Sodium and alcohol intake
- The kidneys are the major regulator of the body's water content
 - They filter excess ions from blood, which then get mixed with water to form urine
 - Healthy = 1-2 quarts of urine a day, but could be more. Gets less efficient when water intake is under 2 cups a day
 - Also remove waste products from bloodstream, such as urea, which sometimes form into kidney stones
- Antidiuretic hormone (ADH) and aldosterone are two hormones that impact fluid balance. The pituitary gland releases ADH in response to dehydration to tell kidneys to conserve water. Adrenal glands secrete aldosterone reduce the elimination of sodium in urine
- Kidneys also secrete enzyme renin to regulate blood pressure via vasoconstriction by stimulating the release of aldosterone

Minerals: the basics

- Major minerals = 100+ g/day
 - Calcium, phosphorus, sodium, potassium, magnesium, chloride, sulfur

TABLE 11.4 Summary of Major Minerals

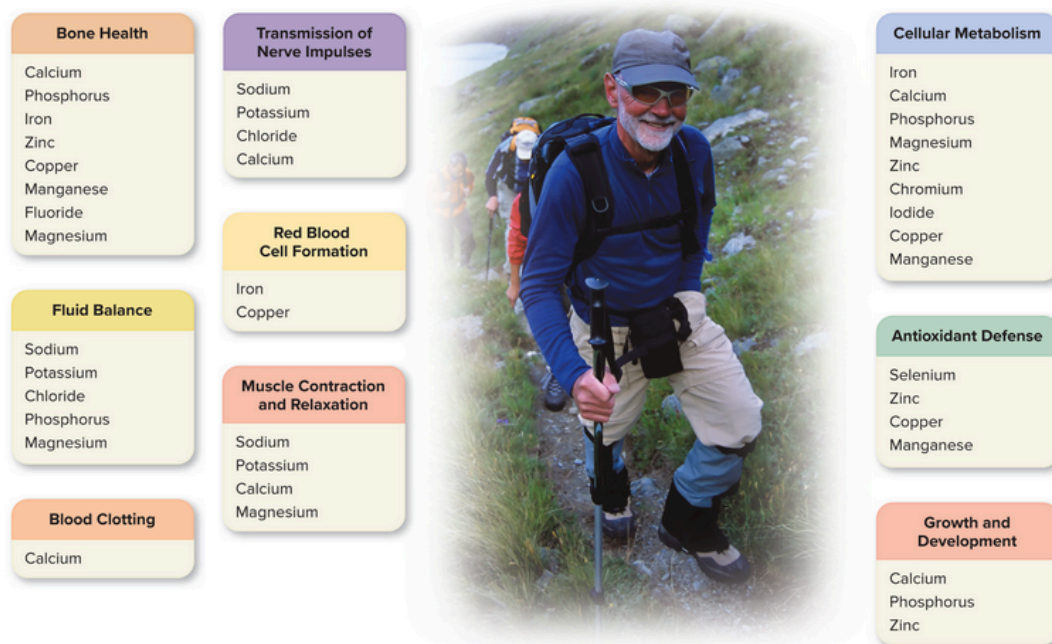
Mineral	Major Functions in the Body	Adult RDA/AI (Adult RDA = Bold)	Major Dietary Sources	Major Deficiency Signs and Symptoms	Major Toxicity Signs and Symptoms
Calcium (Ca)	• Structural component of bones and teeth • Blood clotting • Transmission of nerve impulses • Muscle contraction • Regulation of metabolism	1000–1200 mg	Cow's milk and dairy products, canned fish, leafy vegetables, calcium-fortified foods such as orange juice	• Increased risk of osteoporosis • May increase risk of hypertension	Adult Upper Intake Level (UL)= 2.0–2.5 g/day • Intakes > 2.5 g/day may cause kidney stones and interfere with absorption of other minerals.
Sodium (Na)	• Maintenance of proper fluid balance • Transmission of nerve impulses • Muscle contraction • Transport of certain substances into cells	1500 mg	Table salt; luncheon meats; processed foods; pretzels, chips, and other snack foods; condiments; sauces	• Muscle cramps	No UL has been determined.
Potassium (K)	• Maintenance of proper fluid balance • Transmission of nerve impulses • Maintenance of acid-base balance	2600–3400 mg	Fruits, vegetables, cow's milk, meat, legumes, whole grains	• Irregular heartbeat • Muscle cramps	No UL has been determined.
Magnesium (Mg)	• Strengthens bone • Cofactor for certain enzymes • Heart and nerve functioning	Male: 400–420 mg Female: 310–320 mg	Wheat bran, green vegetables, nuts, chocolate, legumes	• Muscle weakness and pain • Poor heart function	UL = 350 mg/day medication-related • Diarrhea
Phosphorus (P)	• Structural component of bones and teeth • Maintenance of acid-base balance • Component of DNA, phospholipids, and other organic compounds	700 mg	Cow's milk and dairy products, processed foods, soft drinks, fish, baked goods, meat	• None reported	UL = 4 g/day • Poor bone mineralization
Chloride (Cl)	• Maintenance of proper fluid balance • Production of stomach acid • Transmission of nerve impulses • Maintenance of acid-base balance	2300 mg (19–50 years of age)	Processed foods, salty snacks, table salt	• Convulsions (observed in infants)	UL = 3600 mg/day • Hypertension (because of the association with sodium in sodium chloride [table salt])
Sulfur (S)	• Component of organic compounds such as certain amino acids and vitamins	None	Protein-rich foods	• None reported	• Unlikely from dietary sources

- Trace = < 100 g/day

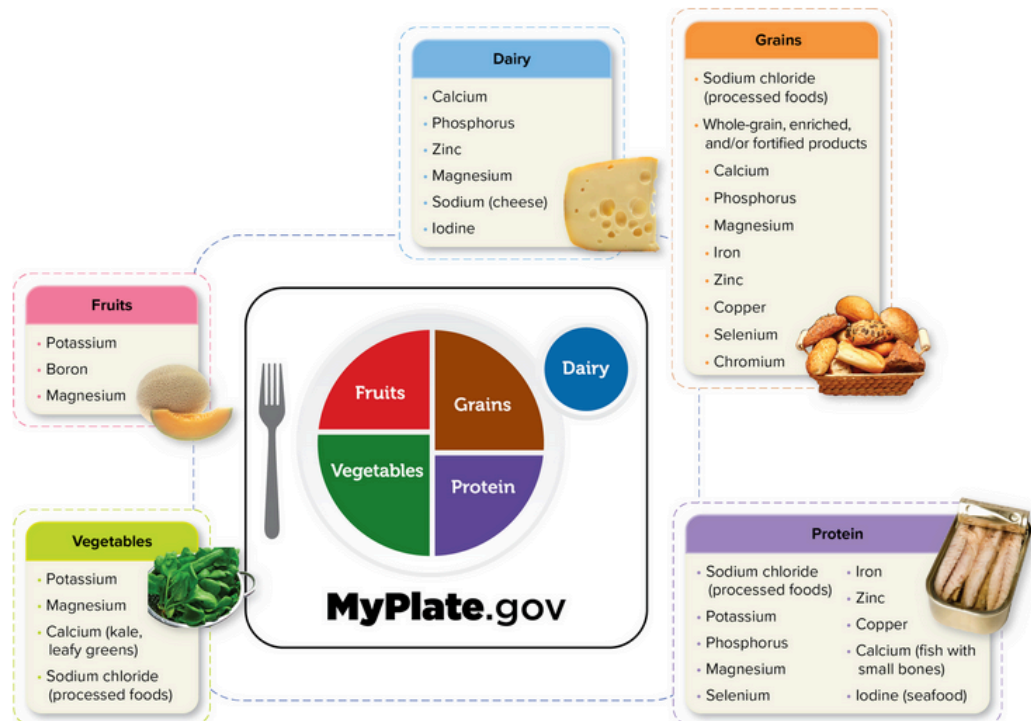
TABLE 11.3 Minerals with Known or Possible Roles in the Body*

Major Minerals	Trace Minerals	Possible Essential Minerals
Calcium (Ca)	Chromium (Cr)	Arsenic (As)
Chloride (Cl)	Copper (Cu)	Boron (B)
Magnesium (Mg)	Fluoride (F) [†]	Lithium (Li)
Phosphorus (P)	Iodine (I)	Nickel (Ni)
Potassium (K)	Iron (Fe)	Silicon (Si)
Sodium (Na)	Manganese (Mn)	Vanadium (V)
Sulfur (S)	Molybdenum (Mo)	
	Selenium (Se)	
	Zinc (Zn)	

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- Minerals cannot be destroyed, unlike vitamins that can be damaged by food prep
 - They are water soluble though, so they can be leached out into cooking water



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- Minerals also have different levels of bioavailability (increase during periods of growth)
- Food sources of minerals:



- Animal foods tend to be better sources, they do not contain phytic or oxalic acid like plants do (interferes with absorption)
 - Plants also lose mineral contents during refinement
- Hard water and supplements are also sources of minerals

Calcium: heart disease and weight control

Calcium rich diets = decrease risk of hypertension, insulin resistance, chronic inflammation, abnormal blood lipid levels, heart disease*

*Calcium supplements increase risk of cardiovascular disease

Dietary calcium also associated with weight loss and improved body composition, but needs more research.

Sodium: hypertension

- Increased risk of CVD/heart disease/ stroke/ kidney failure/ organ damage
- Blood pressure is measured with two values:
 - Systolic pressure (max pressure in an artery)
 - Diastolic pressure (pressure when ventricles relax between heart contractions)
 - Healthy is less than 120/80

TABLE 11.11 Categories for Blood Pressure Levels in Adults (Ages 18 and Older)

Blood Pressure Levels (Adults)			
	Systolic (mm Hg)		Diastolic (mm Hg)
Normal	< 120	and	< 80
Elevated	120–129	and	< 80
Hypertension Stage 1	130–139	or	80–89
Hypertension Stage 2	≥ 140	or	≥ 90
Hypertensive Crisis	> 140	and/or	> 120

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- Major risk factors
 - Family history
 - Advanced age
 - African american ancestry
 - Obesity
 - Physical inactivity
 - Smoking
 - Alcoholism
 - High sodium intake
- Increases as a person ages, partially due to the stiffening of blood vessels